DO NOW — RECALL

Writing, following and documenting algorithms

Name: _____ Class: _____ Date: _____

1. Give a test for a range 0–100 that should be rejected.

2. What does return do?

3. What is the last index for an array of length 4?

4. Why is a[0] unsafe for an empty array?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

−1 or 101.

2. Model answer

Passes a function's result back to the caller.

3. Model answer

3.

4. Model answer

Index 0 does not exist, causing an out-of-range error.

Vocabulary — know these terms

algorithm — Precise finite steps that solve a problem.

precondition — Requirement inputs must meet before execution.

postcondition — Condition guaranteed if the algorithm completes correctly.

invariant — Property maintained during repeated steps.

documentation — Clear description of purpose, inputs and expected behaviour.

NEW LEARNING

Read, annotate and explain

State the precondition

The array must not be empty, or a[0] cannot be read.

Trace the maximum

Set best to 3; compare 7, update best to 7, then ignore 2 and 5.

Document contract

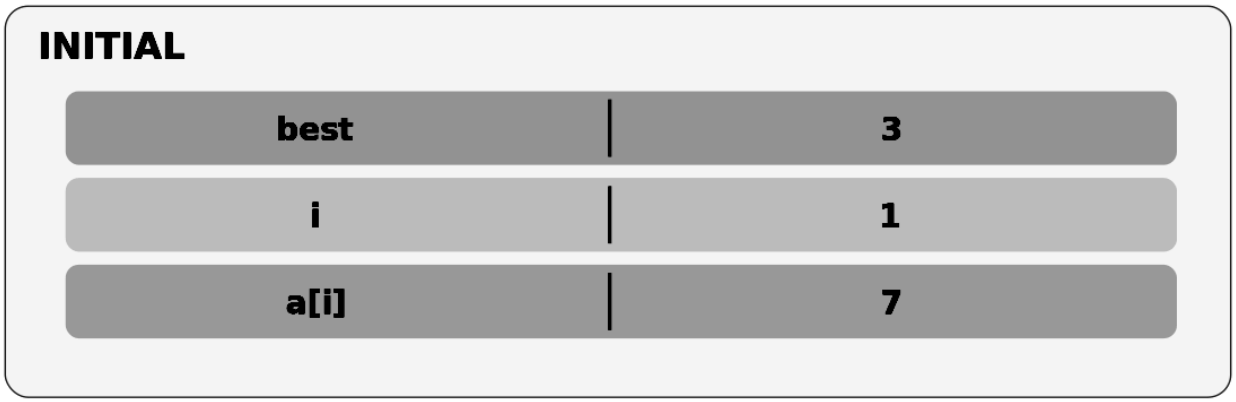
Input: nonempty int array. Output: greatest element. Does not modify input.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

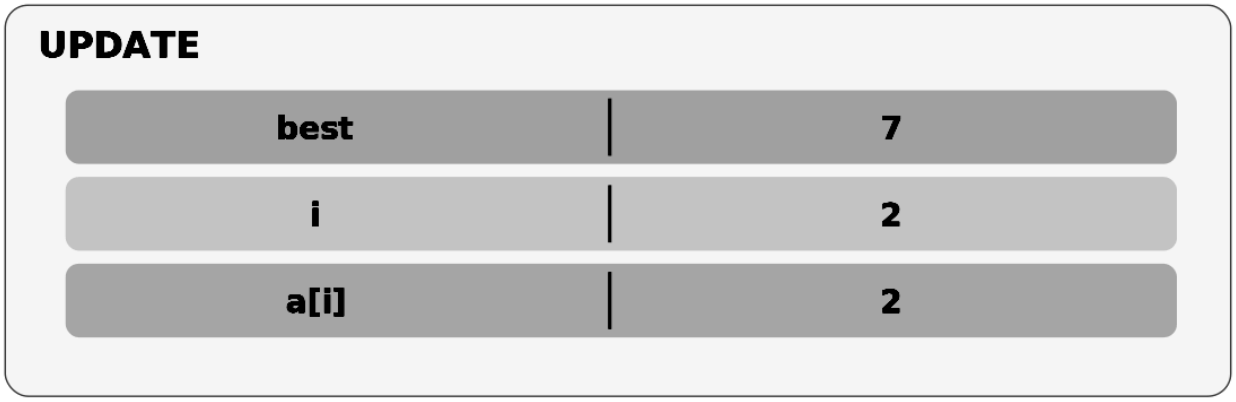
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



INITIAL: best 3; i 1; a[i] 7

New Learning

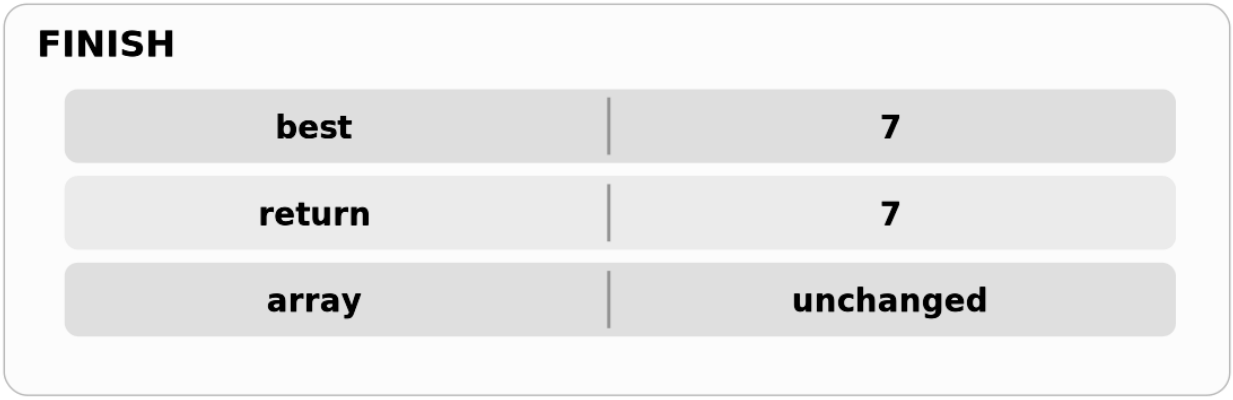


UPDATE: best 7; i 2; a[i] 2

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



FINISH: best 7; return 7; array unchanged

I DO — WORKED EXAMPLE

Follow the worked method

Trace Largest([4,2,9]) and write the returned result.

Predict first

Start best at first element 4; 2 does not replace it.

Model and explain

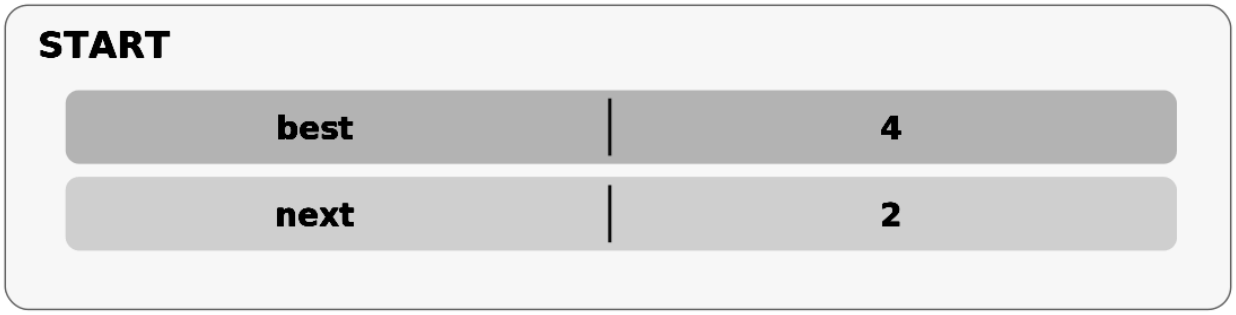
9 is larger so best becomes 9; return 9.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace Largest([4,2,9]) and write the returned result.

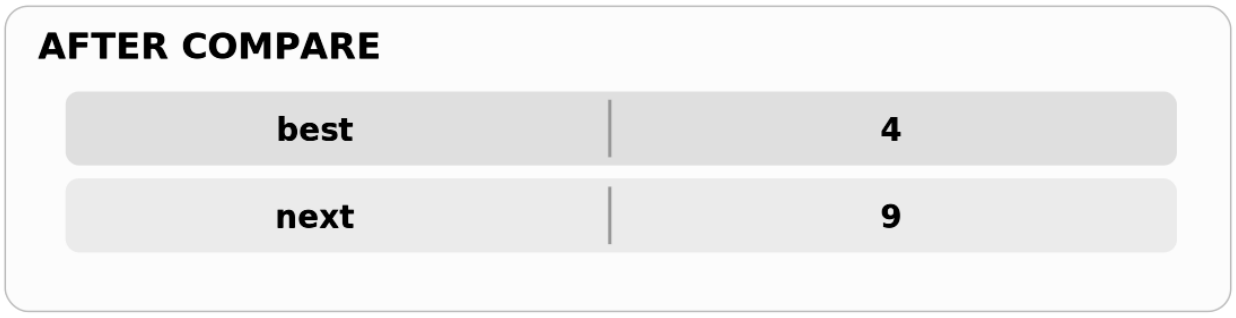
Start best at first element 4; 2 does not replace it.



START: best 4; next 2

Trace Largest([4,2,9]) and write the returned result.

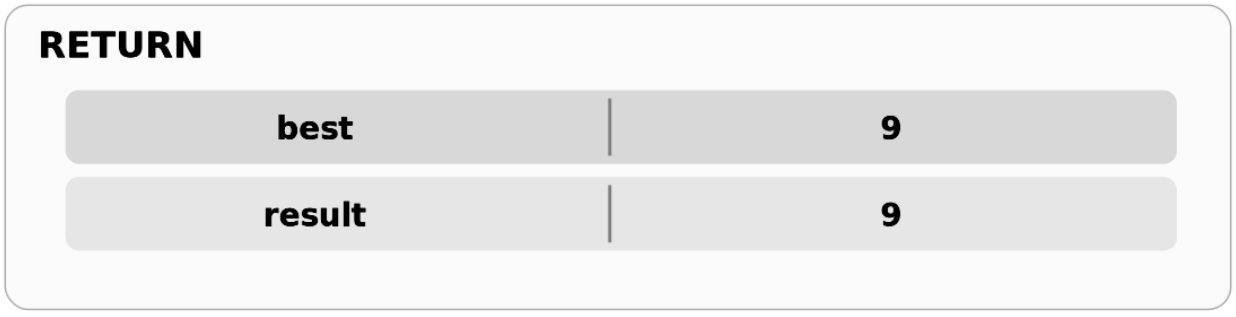
Start best at first element 4; 2 does not replace it.



AFTER COMPARE: best 4; next 9

Trace Largest([4,2,9]) and write the returned result.

Start best at first element 4; 2 does not replace it.



RETURN: best 9; result 9

WE DO — GUIDED PRACTICE

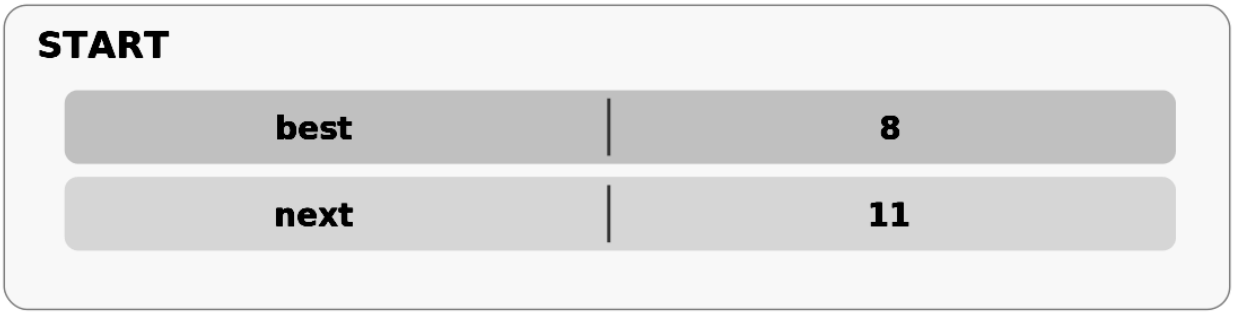
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace Largest([8,11,6,10]) and state its precondition.

11 replaces initial best 8; neither 6 nor 10 exceeds 11.



START: best 8; next 11

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first 11 replaces initial best 8; neither 6 nor 10 exceeds 11.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace Largest([2,5,1]) and state initial and final best.

Question 2

Write pseudocode to return the minimum of a nonempty array values and explain its precondition.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate what happens for Largest([-4,-2,-9]) and explain why initialising best=0 would be incorrect.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Initialising a maximum-tracking variable to 0 is correct for every integer array.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

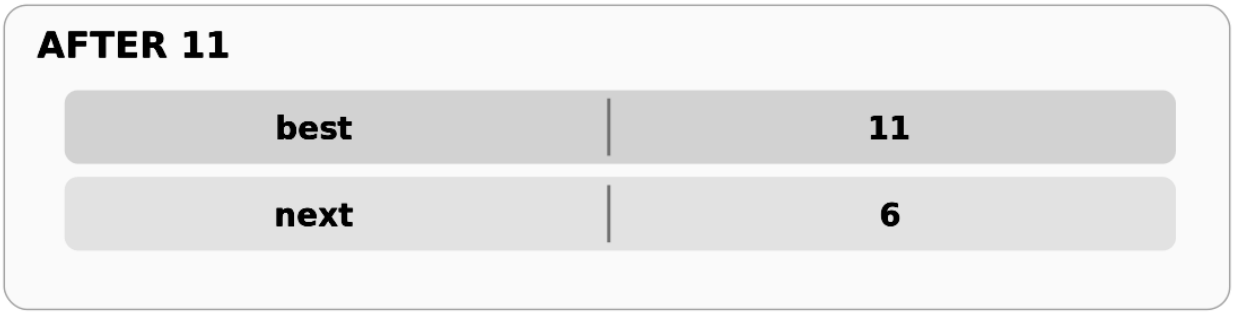
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace Largest([8,11,6,10]) and state its precondition.

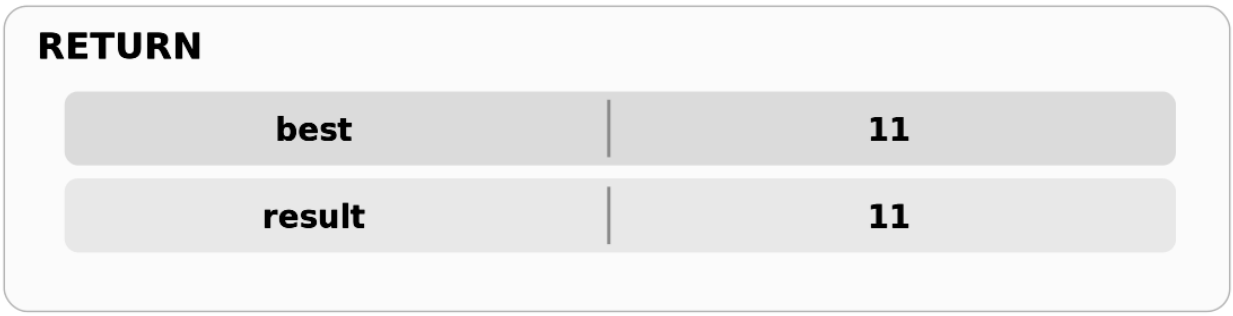
11 replaces initial best 8; neither 6 nor 10 exceeds 11.



AFTER 11: best 11; next 6

Trace Largest([8,11,6,10]) and state its precondition.

11 replaces initial best 8; neither 6 nor 10 exceeds 11.



RETURN: best 11; result 11

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and explain

The output is 11; the required precondition is a nonempty array.

IMPROVE: Explain one correction to your own reasoning.